

FAA Rules for Drone Pilots

Week 3 — Tuesday Lecture

Same drone, two flights. Same rules?

A friend flies it at the park for fun. You fly it to photograph a client's roof for a repair estimate.

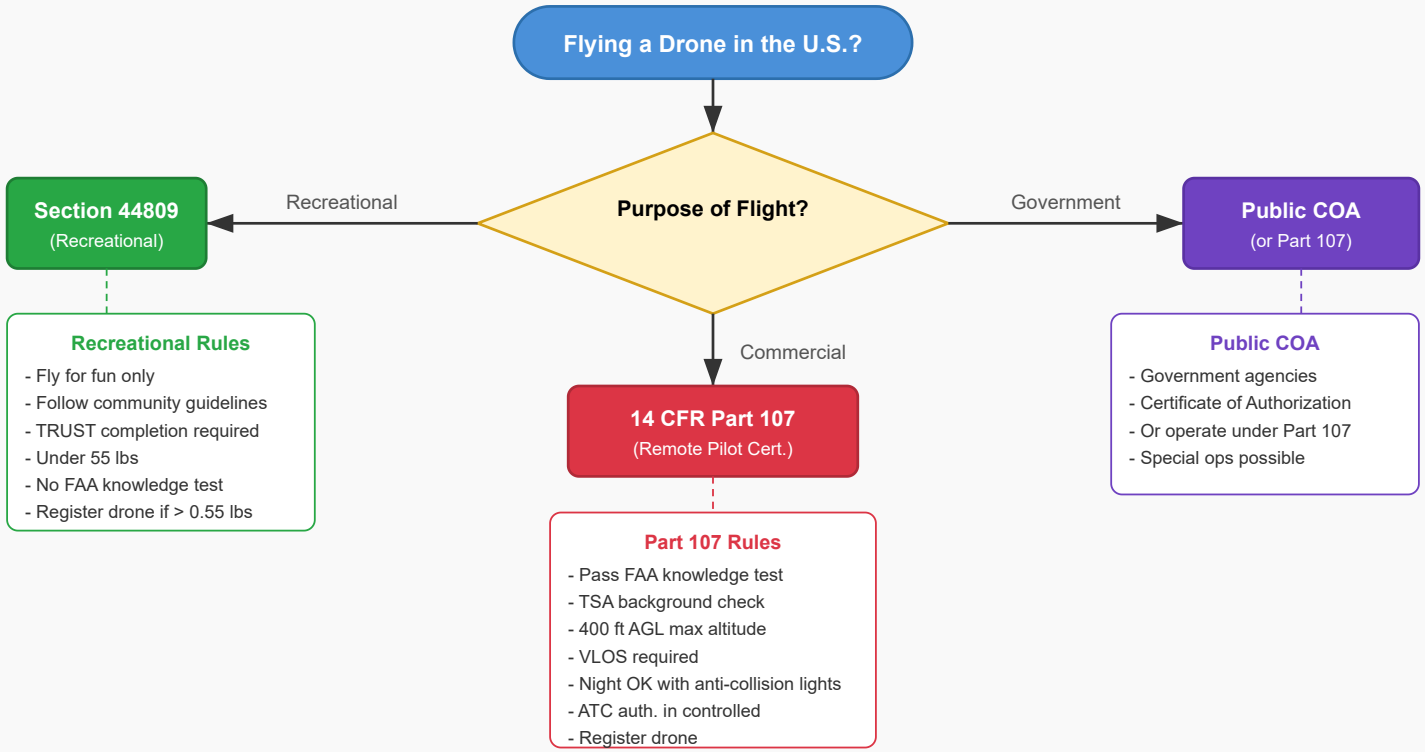
ANSWER No — the purpose decides the rules

- **Recreational** flying: the exception in 49 U.S.C. 44809, with the **TRUST** certificate
- **Anything else** — a job, a client, a class mapping project, research: **Part 107**
- Part 107 needs a **remote pilot certificate** and an aircraft registered for Part 107
- In this class, a **TA with a Part 107 certificate** is remote pilot in command

Today

- Which rules apply to your flight
- The operating limits
- Airspace: where you need permission
- People, crowds and moving vehicles
- The paperwork: registration, Remote ID, reporting
- Getting your certificate
- When a flight goes wrong

Figure 1 — UAS Operations Regulatory Pathway

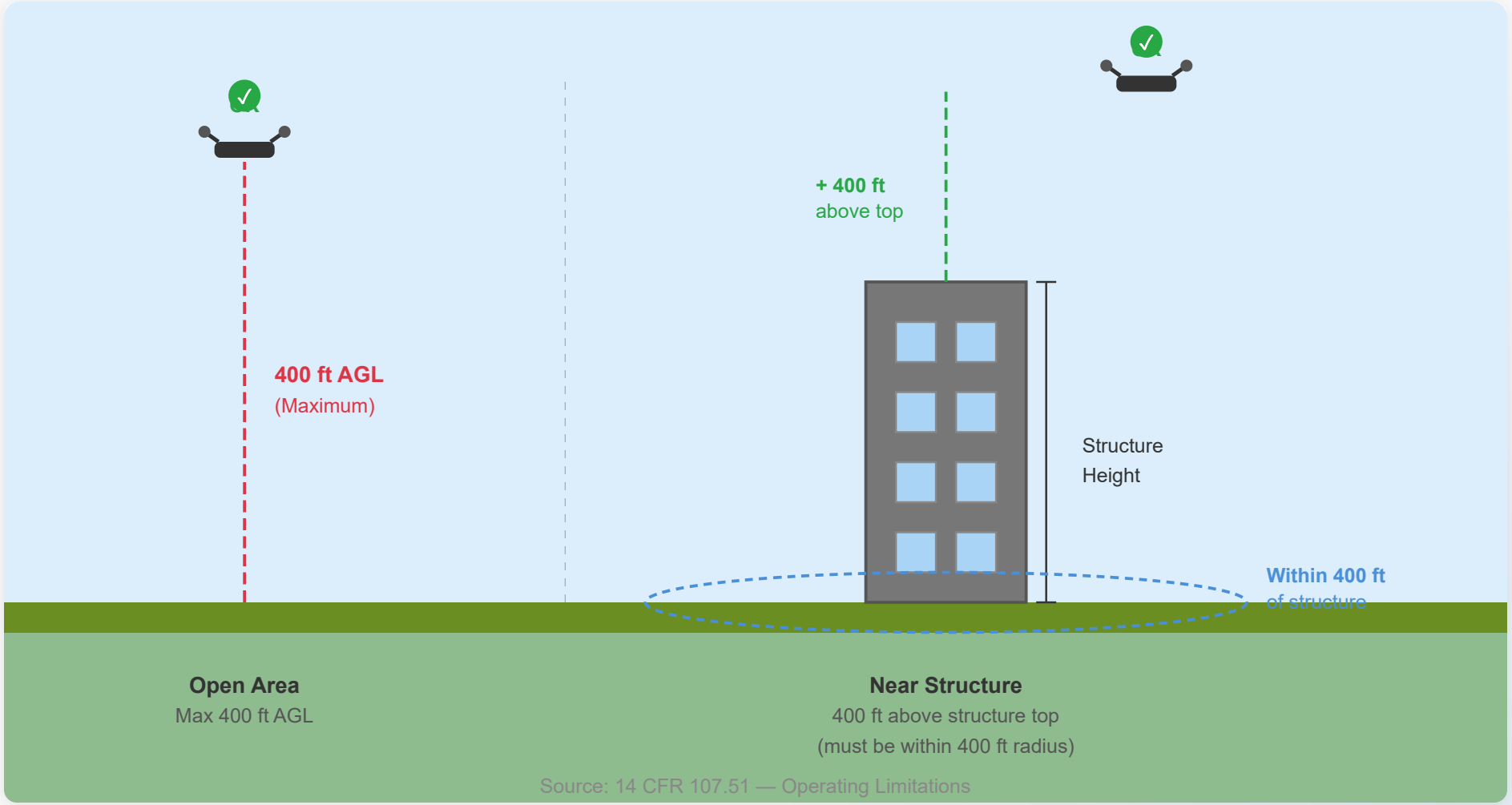


Source: 14 CFR Part 107, Section 44809, FAA UAS Regulations

Start with the purpose of the flight. That decides which set of rules you fly under.

The operating limits (§107.51 and others)

Limit	Value
Altitude	400 ft above ground level (AGL), or within 400 ft of a structure
Speed	87 knots (100 mph)
Visibility	3 statute miles from the control station
Clouds	500 ft below, 2,000 ft horizontally
Sight	Visual line of sight — yours or a visual observer's
Right of way	Yield to every other aircraft
Alcohol	Not within 8 hours, and under 0.04 %



400 ft above the ground — or up to 400 ft above the top of a structure, if you stay within 400 ft of it.

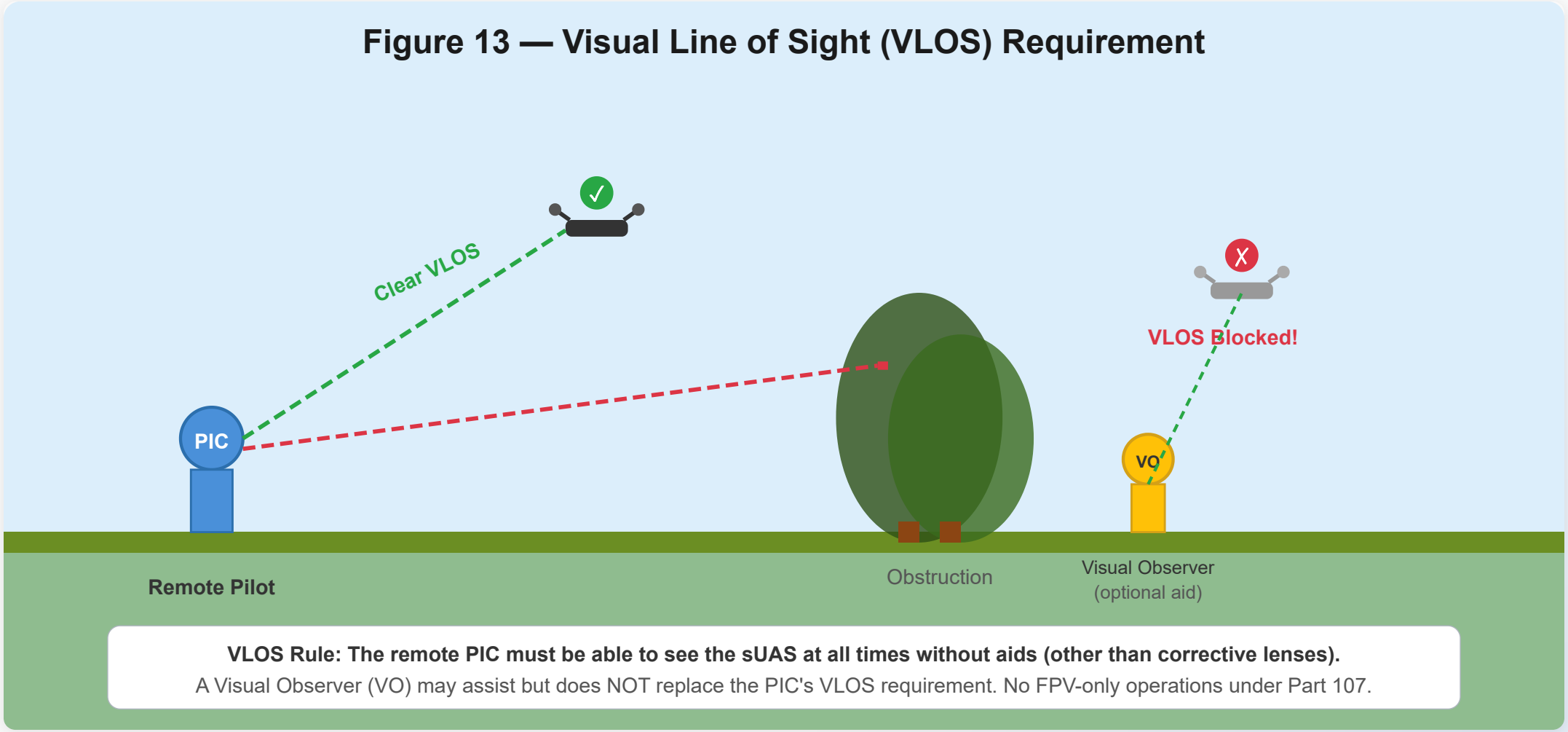
A tower is 300 ft tall. Can you inspect its top at 650 ft AGL?

A) No, never above 400 ft · B) Yes, if you stay within 400 ft of the tower · C) Only with a waiver

ANSWER B — the structure exception

- Within **400 ft horizontally** of a structure, you may fly up to **400 ft above its top**
- 300 ft tower + 400 ft = **700 ft AGL** allowed, so 650 ft is fine
- Step away from the tower and the 400 ft AGL limit applies again
- An inspection flight is exactly what the exception is for

Figure 13 — Visual Line of Sight (VLOS) Requirement



Visual line of sight: you, or a visual observer beside you, can see the aircraft with your own eyes the whole time.

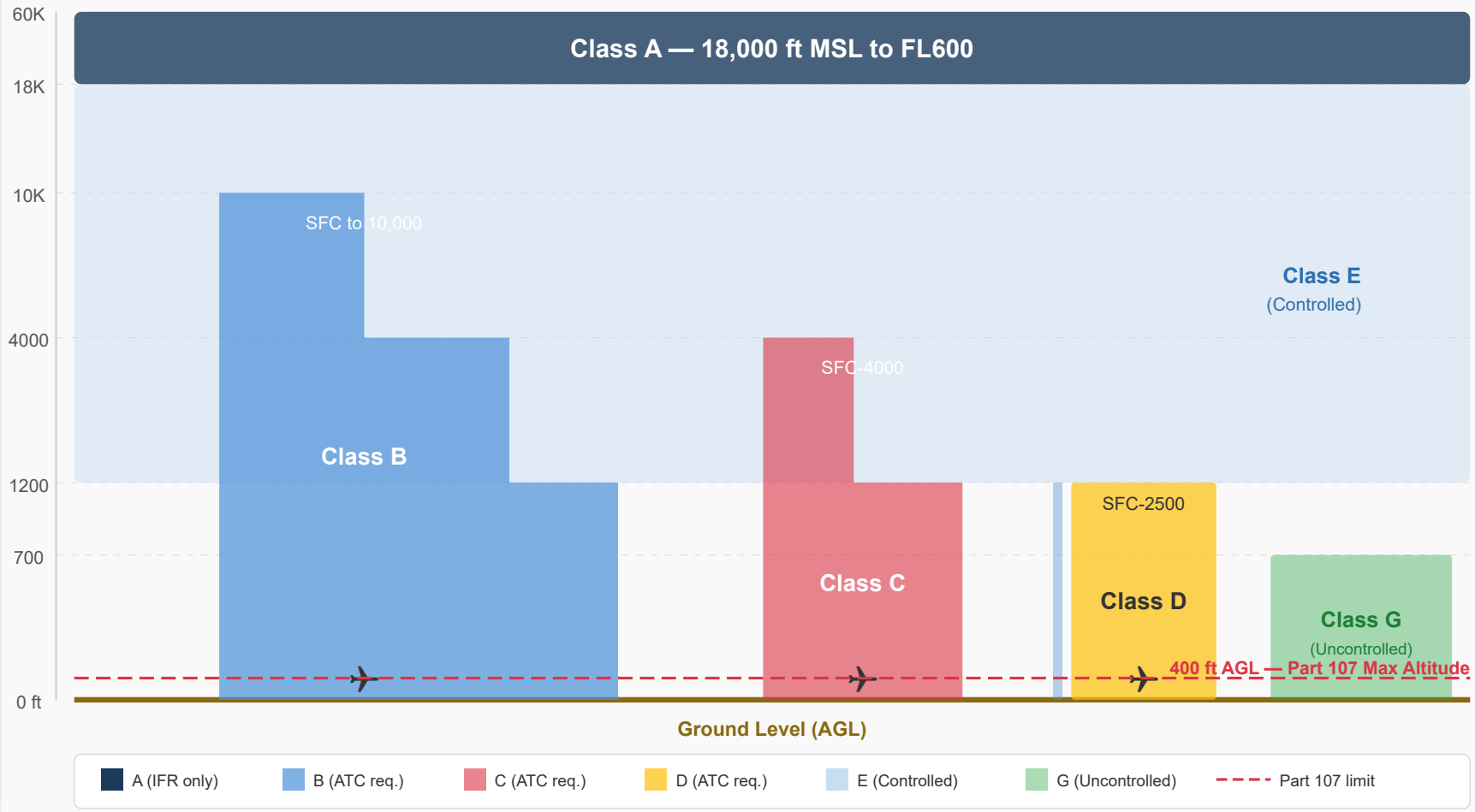
Sunset was 20 minutes ago. Can you finish the flight?

A) No — Part 107 is daylight only · B) Yes, with anti-collision lighting · C) Only with a night waiver

ANSWER B — night flying is allowed

- Since **2021**, night operations need no waiver (§107.29)
- The pilot must have passed the current test or training
- **Anti-collision lights visible for 3 statute miles**, flashing enough to avoid a collision
- The same lights are required in **civil twilight** — 30 minutes before sunrise and after sunset

Figure 11 — National Airspace System (NAS) Classification



Airspace classes, stacked. Controlled airspace needs permission before you fly.

Where you need permission

- **Controlled airspace** — Class B, C, D and **surface** Class E: ATC authorization (§107.41)
- **LAANC**: authorization in seconds, through an FAA-approved app, up to a set ceiling
- **FAA DroneZone**: everything LAANC can't do, which takes longer
- **Restricted areas**: permission from the controlling agency
- **Temporary flight restrictions (TFRs)**: check for them before every flight
- **BYU campus is inside Provo's Class D** — every campus flight needs LAANC *and* a BYU permit

A home football game at LaVell Edwards Stadium.

A TA with a Part 107 certificate wants to film it from the air. Can they?

ANSWER No — a standing flight restriction covers the stadium

- FAA notice **FDC 0/0367**: no manned or unmanned aircraft within **3 NM** and **3,000 ft AGL** of a stadium with **30,000+ seats**
- Applies to NFL, MLB, **NCAA Division I football** and major races, **1 hour before to 1 hour after**
- LaVell Edwards Stadium seats about **62,000** — every home game qualifies
- LAANC cannot authorize it; only the event, security or the broadcaster can get a waiver
- Fines up to **$75,000** per violation, and drones can be seized

Figure 15b: Flying over people, the four categories

§107.39 and Subpart D. No category is needed over people taking part, under a covered structure, or inside a stationary vehicle.

Category 1



0.55 lb or less

takeoff weight, everything on board

No Declaration of Compliance needed. No exposed rotating parts that could cut skin.

Weigh it with everything attached: lights, guards, payload.

§107.110

Category 2



under 11 ft·lb

impact energy on a person

FAA-accepted Declaration of Compliance from the maker, and a category label on the aircraft. No exposed parts that could cut skin.

§107.115, §107.120

Category 3



under 25 ft·lb

impact energy on a person

Declaration and label, plus: only over a closed or restricted site with people on notice, or no sustained flight over anyone not under cover.
Never over open-air assemblies.

§107.125, §107.130

Category 4



Airworthiness certificate

issued under Part 21

Certified like a manned aircraft: flown to its Flight Manual limits, maintained and inspected to its manual, records kept.

§107.140

Declaration of Compliance: the manufacturer declares it, the FAA accepts it, the label on the aircraft shows the category.

The pilot's job is to check the label. This is the most-missed item on the knowledge test (FAA statistics, 2023 to 2025).

Sustained flight over an open-air assembly (a crowd): Categories 1, 2 and 4 only with Remote ID (§89.110 or §89.115(a)); Category 3 never (§107.125).

Flying over people depends on the aircraft's category. The category is printed on the aircraft's label.

Your 0.6 lb drone has no Declaration of Compliance.

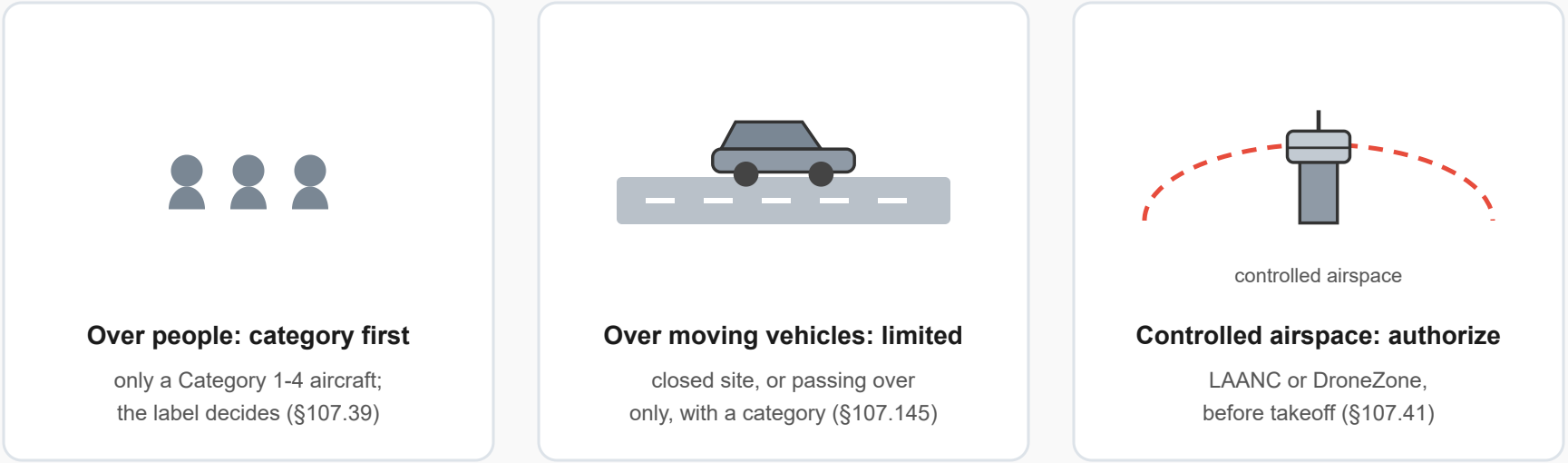
Can you fly it over people at an outdoor graduation? A) Yes, it is light · B) No · C) Only if they agree

ANSWER B — over 0.55 lb needs a declared category

- **Category 1** is 0.55 lb or less, with no exposed parts that could cut skin
- Heavier aircraft need an FAA-accepted **Declaration of Compliance** for Category 2 or 3
- The **manufacturer** declares it; the **pilot** checks the label
- Category 3 may never fly over an open-air crowd; sustained flight over crowds needs Remote ID
- Moving vehicles have their own limits (§107.145)

Figure 12: Three things to settle before takeoff

None of them can be fixed in the air: check the category, what is below the route, and the airspace before you launch.



Over people: category first

only a Category 1-4 aircraft;
the label decides (§107.39)

Over moving vehicles: limited

closed site, or passing over
only, with a category (§107.145)

Controlled airspace: authorize

LAANC or DroneZone,
before takeoff (§107.41)

Three things to settle before takeoff. None of them can be fixed in the air.

Figure 15: Which rules can be waived?

A certificate of waiver (§107.200) can relax only the ten rules listed in §107.205. Everything else in Part 107 stands as written.

Can be waived (§107.205)

Flying from a moving vehicle or aircraft	§107.25
Lights at night and civil twilight	§107.29(a)(2), (b)
Visual line of sight	§107.31
Visual observer	§107.33
One pilot flying several aircraft	§107.35
Yielding the right of way	§107.37(a)
Flying over people	§107.39
Flying in controlled airspace	§107.41
Altitude, speed, visibility, cloud clearance	§107.51
Flying over moving vehicles	§107.145

Cannot be waived

Careless or reckless flying; dropping things	§107.23
Alcohol or drugs	§107.27
Carrying hazardous material	§107.36
Transponder or ADS-B Out switched on	§107.52, .53
Registering the aircraft	§107.13

…and every other section that is not on the left. The FAA has no form for those, so there is nothing to apply for.

Even a waiver never allows carrying someone else's property for hire beyond line of sight or from a moving vehicle (§107.25, §107.31).

Night flying needs no waiver since April 2021: anti-collision lights visible 3 statute miles and current training (§107.29). Apply for a waiver through FAA DroneZone with a safety case; the Part 107 waiver is the exam's name for that paperwork.

Some limits can be waived by the FAA for a specific operation. The rest cannot.

The paperwork

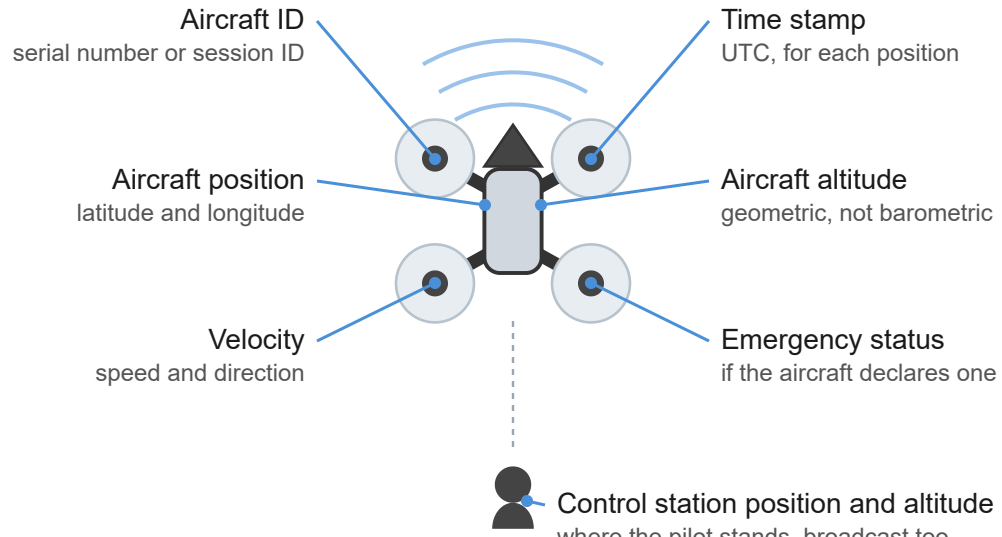
- **Register** every Part 107 aircraft, whatever it weighs, and mark the number on it
- **Remote ID**: three ways to comply
- **Carry** your certificate and photo ID; show them when asked (§107.7)
- **Preflight**: weather, airspace, NOTAMs and TFRs, aircraft condition, crew briefing

Figure 27: Three ways to comply with Remote ID (Part 89)

Required since 16 September 2023 for any drone that has to be registered. §89.110 built in; §89.115(a) add-on module; §89.115(b) FRIA.

1. Standard Remote ID, built into the aircraft (§89.110)

broadcasts these message elements from takeoff to shutdown (§89.305)



2. Broadcast module (§89.115(a))

An add-on for a drone without built-in Remote ID. It broadcasts the aircraft's ID, position, altitude and velocity, but the takeoff location instead of the control station (§89.315).
Keep the aircraft in sight at all times; check the module before takeoff.

3. Fly inside a FRIA (§89.115(b))

An FAA-Recognized Identification Area, usually a model-club or school field.
No Remote ID equipment is needed, but the aircraft and the pilot both stay inside the area, and the aircraft stays in sight.

Not an option on a job site: a FRIA is that one field.

If the broadcast stops in flight, land as soon as practicable (§89.110(a)(2), §89.115(a)(4)).

The serial number of the aircraft or the module goes on the Certificate of Aircraft Registration (§89.110(b)(4), §89.115(a)(1)).

Remote ID: the aircraft broadcasts who and where it is, so others can identify it.

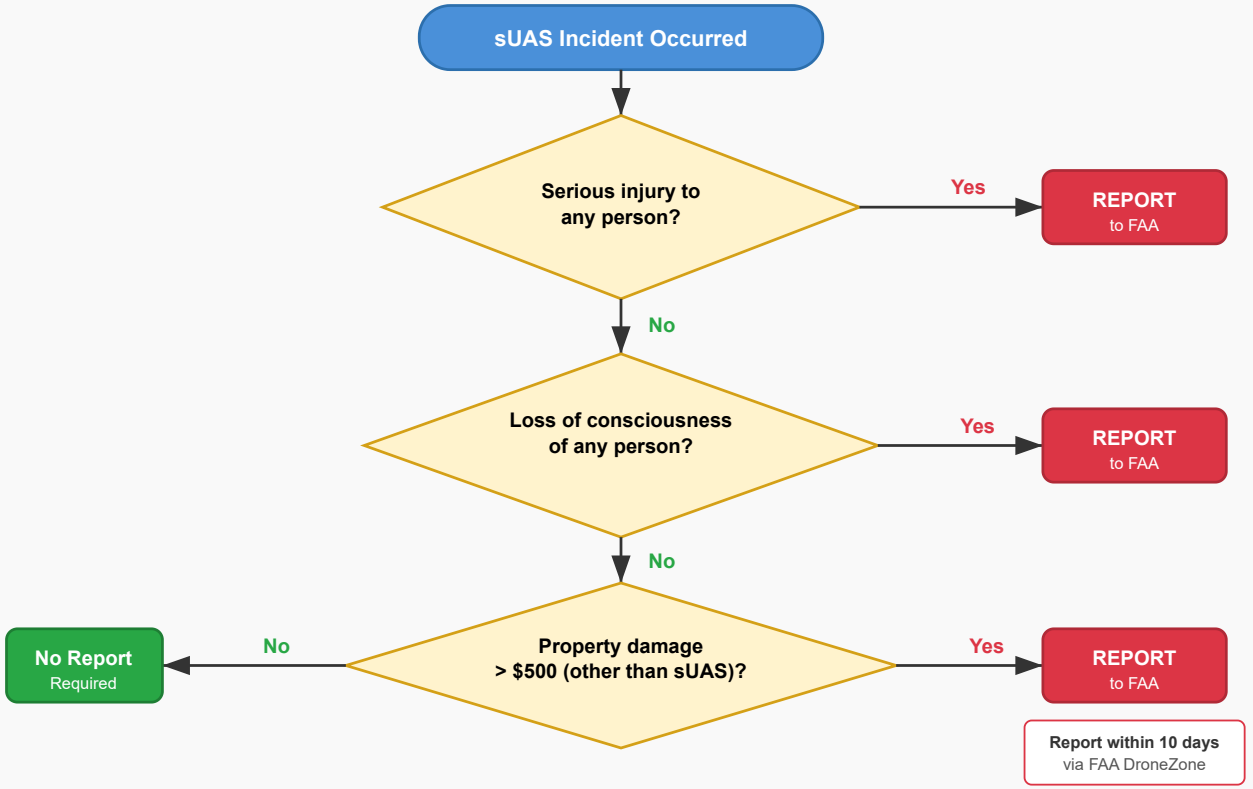
The Remote ID warning comes on mid-flight.

What do you do? A) Keep flying, it's only a broadcast · B) Land as soon as practicable · C) Climb for a better signal

ANSWER B — land as soon as practicable

- A standard Remote ID aircraft that stops broadcasting must **land as soon as practicable** (§89.110)
- Without Remote ID the flight is not legal, unless you are inside an FAA-recognized area
- Most aircraft also refuse to take off with Remote ID failed

Figure 26 — sUAS Accident Reporting Decision Tree



Not every mishap is reportable. Serious injury or more than $500 of property damage is.

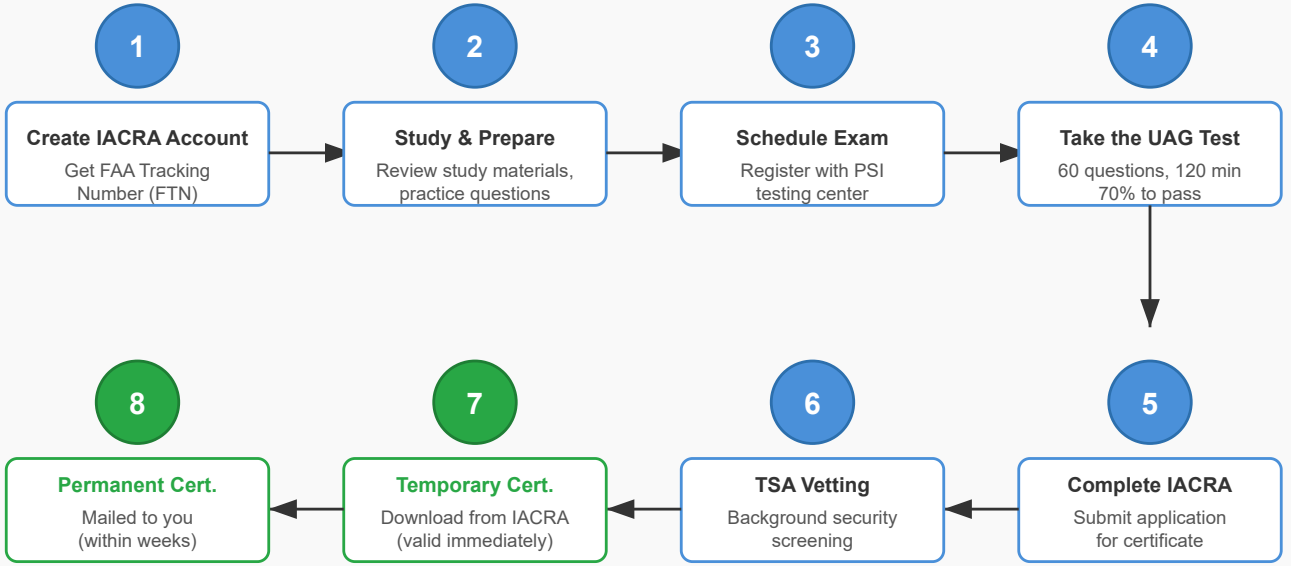
You clip a parked car. The mirror costs $650 to fix.

Who do you tell, and how soon?

ANSWER The FAA, within 10 days

- Report to the **FAA within 10 calendar days** (§107.9)
- Reportable: **serious injury**, **loss of consciousness**, or **property damage over $500** (repair or value)
- The aircraft itself does not count toward the $500
- Also exchange details with the car's owner, as with any accident

Figure 4 — Part 107 Certification Process



Source: FAA.gov — Become a Drone Pilot

From account to certificate: IACRA, the knowledge test, the application, and a TSA check.

Getting your certificate

1. Create an **IACRA** account and get your FAA Tracking Number (FTN)
2. Schedule the test with **PSI** — locally at UVU's testing centre at the Provo airport
3. Pass: 60 questions, 2 hours, **70 %**. *As of October 2026 the fee was about $175; check PSI when you book.*
4. Apply in IACRA with your exam ID; the **TSA** checks your background
5. Print the **temporary certificate**; the plastic card comes by mail

Week 9 is all about passing that test.

Figure 25: Staying current

The certificate never expires. The right to use it does: recurrent training every 24 calendar months (§107.65).



Carry the certificate and a photo ID whenever you fly (§107.7).

Show them on request to the FAA, the NTSB, the TSA or any federal, state or local law enforcement officer.

The certificate does not expire. To keep flying, take the free online recurrent training every 24 calendar months.

When a flight goes wrong

- **Lost link**: the aircraft should return home on its own — know where "home" is set
- **Fly-away**: keep your eyes on it, warn people nearby, note where it goes
- **Low battery**: land early; cold days drain batteries faster
- **Battery fire or swelling**: don't fly it; move it outdoors onto a non-flammable surface
- Details: Common Flight Issues

Your aircraft stops responding and starts drifting away.

What are the first three things you do?

ANSWER Stay calm, keep sight, warn

1. **Keep your eyes on it** and tell your visual observer
2. **Trigger return-to-home**, or try to regain the link
3. **Warn people** where it is heading; note where it lands

- Afterwards: report it if it caused injury or more than $500 of damage

What to do Thursday

- **Flight Practice lab:** Flight Practice — Holy Stone mini drones, indoors
- A TA is remote pilot in command for every class flight
- Read before Thursday: FAA Part 107 Overview · Common Flight Issues
- Week 9 returns to these rules as exam practice